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****LAB TASK 06****

Data Visualization in Excel

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Topic: Overlaid Gridlines Chart

Objective :

The objective of this lab is to learn how to create **advanced Excel visualizations using Overlaid Gridlines**. This technique helps viewers analyze values easily by showing **reference grid lines over column charts**.

These gridlines make it easier to compare values across different categories.

Dataset Description

The dataset used in this lab represents **regional sales data** along with additional reference values used for creating advanced data visualizations such as **Overlaid Gridline Charts** in Microsoft Excel.

The purpose of this dataset is to demonstrate how column charts can be enhanced with additional line or scatter elements to create visual gridlines that improve the readability and interpretability of the chart.

The dataset consists of **5 observations (rows)** and **6 variables (columns)**. Each row represents a different geographical region, while the columns contain sales values and gridline reference values.

Meaning of Data

- **Region** → Different geographical areas
- **Sales (Main Series)** → Actual sales values
- **Grid Values** → Used to create horizontal gridline levels (10,20,30,40)

Task 1 — Create Overlaid Gridline Chart (Line Method)

Step 1 — Create Column Chart

1. Select **Region and Sales columns**
2. Go to
3. **Insert → Column Chart → Clustered Column**

Step 2 — Remove Chart Title

Click the title and press **Delete**.

This keeps the chart clean and professional.

Step 3 — Add Gridline Series

Now add the columns:

Grid_10
Grid_20
Grid_30
Grid_40

Method:

Right Click Chart → Select Data → Add Series

Now the chart becomes **clustered bars with 5 series**.

Step 4 — Convert Grid Series into Lines

1. Click any **Grid series**
2. Go to

Chart Design → Change Chart Type

Change:

Grid_10 → Line
Grid_20 → Line
Grid_30 → Line
Grid_40 → Line

Keep **Sales as Column Chart**.

Now you will see **lines across bars**.

Step 5 — Move Lines to Secondary Axis

1. For each line series:

   Right Click Line → Format Data Series

2. Select:

   Secondary Axis

   A new **axis appears on the right side**.

3. Set both axes range: **0 – 50**

   So both scales match.

Step 6 — Enable Secondary Horizontal Axis

Click chart → press **+ (Chart Elements)**

Enable:

Secondary Horizontal Axis

This allows lines to stretch properly across columns.

Step 7 — Change Line Color

Select all line series and set:

Color → White
Width → 1.5 pt

This creates the **gridline illusion across bars**.

Step 8 — Adjust Axis Position

Right click **Secondary Horizontal Axis**

Change:

Position Axis → On Tick Marks

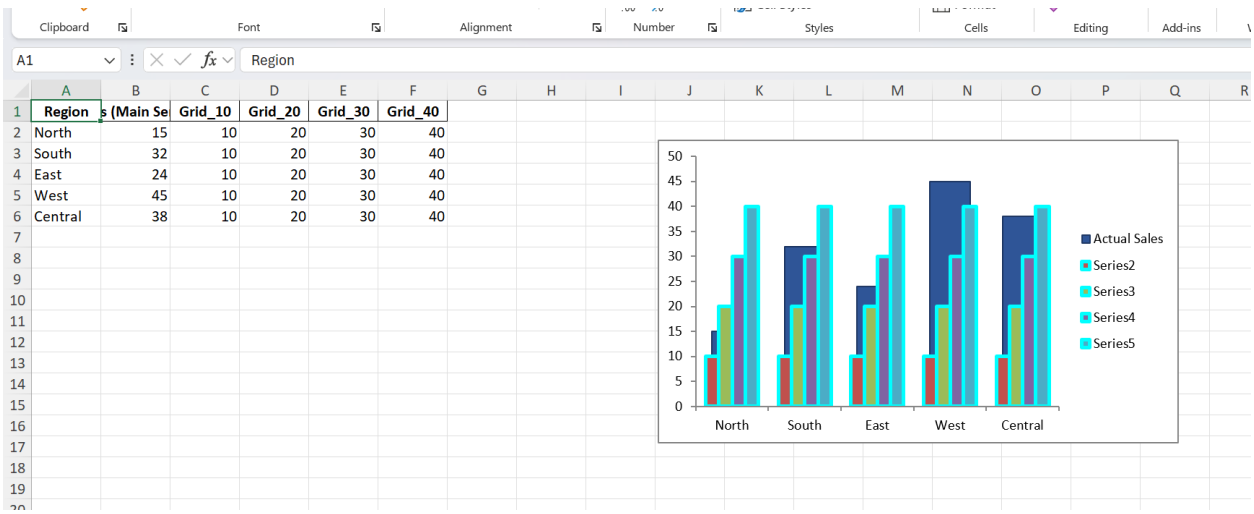
This aligns the lines perfectly across columns.

Step 9 — Hide Secondary Axes

Remove extra axes formatting:

Secondary Horizontal Axis → No line
Tick marks → None
Labels → None

Output (Screenshot) :



Task Description

In this task, an **Overlaid Gridline chart** is created using a combination of a **column chart and multiple line series**. The primary column chart represents the **main sales values for each region**, while additional line series are added to represent different grid levels (10, 20, 30, and 40).Initially, a column chart is created using the main dataset values. Afterward, additional series are inserted into the chart and converted into **line charts**. These lines are placed on a **secondary axis** so they can extend across the entire chart. By adjusting axis settings and formatting the lines in white color, the lines appear as **gridlines over the columns**.This technique helps in creating a visually enhanced chart where the gridlines appear directly on top of the columns, making it easier to interpret and compare values.

Insights from Task 1

1. The **West region shows the highest sales value**, reaching approximately **45 units**, which indicates strong performance in that region.
2. The **Central and South regions also perform well**, with sales values around **38 and 32 respectively**, showing moderate to high performance.
3. The **North region has the lowest sales**, indicating that this region may require improvement strategies.
4. The gridlines at **10, 20, 30, and 40** provide clear reference levels that make it easier to estimate the height of each bar.
5. This visualization technique significantly improves the **readability and professional appearance** of the chart compared to a simple column chart.

Task 2 — Overlaid Gridlines Using Formula Method

This method uses **stacked column charts**.

Steps

1. Select formula data range

   C16 : M20

2. Insert chart

   Insert → Stacked Column Chart

3. Go to

   Chart Design → Switch Row/Column

Now rows become series.

Styling

1. Select each column
2. Change

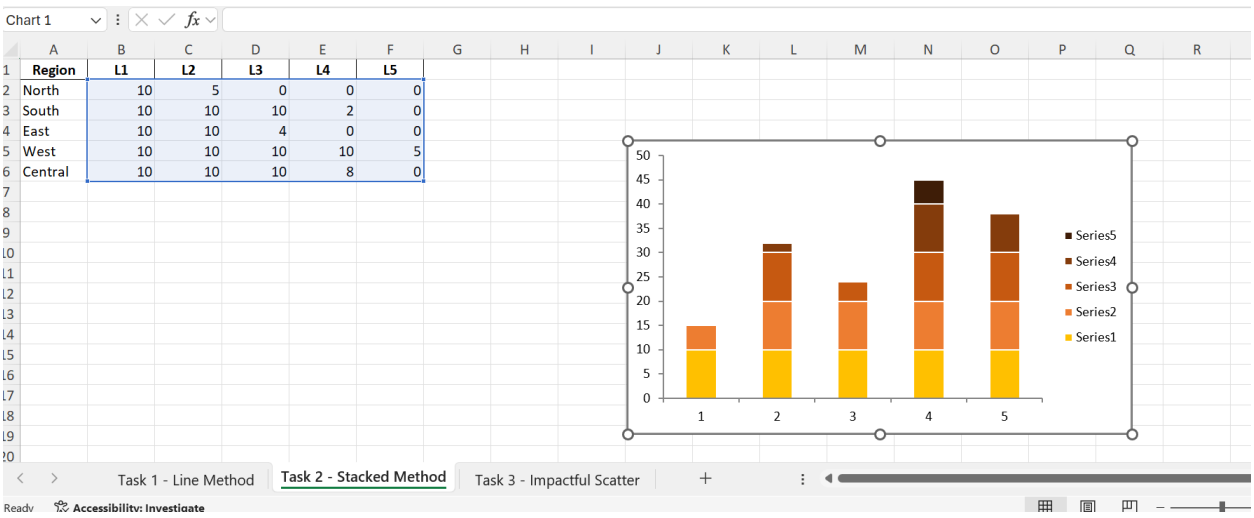
Fill Color → Same Color
Outline → White
Width → 1.5pt

3. Remove

   Gridlines
   Legend
   Title

Now the chart looks like it has **built-in gridlines**.

Output (Screenshot) :



Task Description

In this task, the Overlaid Gridline chart is created using a **stacked column chart combined with formula-generated data**. Instead of manually adding line series, formulas are used to divide the columns into multiple stacked segments that visually resemble gridlines.

First, a **stacked column chart** is created from the formula-generated data. Then the rows and columns of the chart are switched so that the data is displayed correctly. Afterward, formatting is applied to each stacked segment by setting the **fill color to the same shade** and the **outline color to white**.

The white outlines between stacked segments create the illusion of horizontal gridlines across the chart.This approach is slightly more complex in terms of dataset preparation but requires fewer chart modifications.

Insights Task 2

1. The stacked column visualization allows users to clearly observe **how sales values compare against multiple reference levels**.
2. The segmentation of columns makes it easier to visually estimate where each region falls between different levels.
3. This method creates a **clean and structured chart design**, which is often used in professional dashboards.
4. Compared to the first method, this technique relies more on **data structure and formulas rather than chart editing**.
5. The visualization effectively highlights that **West and Central regions consistently exceed the 30-unit level**, indicating stronger performance.

Task 3 — Overlaid Gridlines Using Scatterplot

This method uses **error bars to create gridlines**.

Step 1

Create **column chart for sales data**.

Delete:

Chart title
Default gridlines

Step 2

Add new series called **Scatter**.

Use:

Y values → B10:B14
X values → A10:A14

Step 3

Change chart type:

Scatter series → Scatter Plot

Step 4 — Add Error Bars

Go to:

Chart Elements → Error Bars → More Options

Delete **Vertical Error Bars**.

Keep **Horizontal Error Bars**.

Step 5 — Customize Error Bars

Set:

Direction → Both
End Style → No Cap

Custom error value:

2.4

This extends lines across the chart.

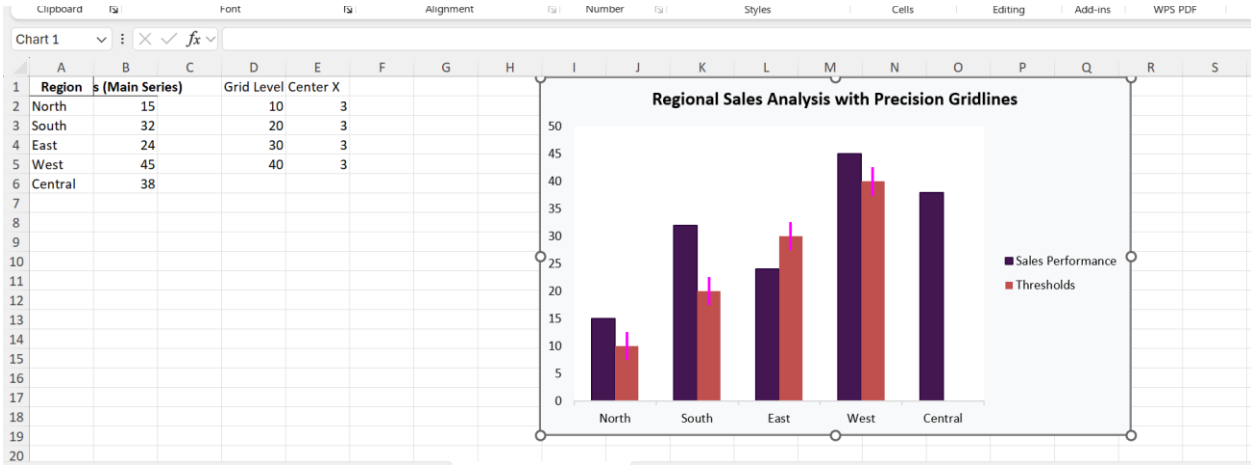
Step 6 — Hide Scatter Points

Format scatter series:

Marker → None

Now only **horizontal lines remain**, creating the gridline effect.

Output (Screenshot):



Task Description

In this task, the Overlaid Gridline effect is created by combining a **column chart with a scatterplot chart**. The scatterplot is used to generate horizontal lines across the chart by adding **error bars**.

First, a standard column chart is created using the main sales data. Then, a scatterplot series is added to the chart using predefined coordinates. Horizontal error bars are applied to the scatterplot points and extended across the width of the chart.

After removing the scatterplot markers, only the horizontal error bars remain visible. These bars act as **gridlines that overlay the column chart**, creating the desired visualization effect.

Insights Task 3

1. The scatterplot method provides a **flexible way to create custom gridlines** in charts.
2. The horizontal error bars serve as **visual reference lines**, helping viewers quickly estimate data values.
3. The technique allows precise control over **gridline placement and length**, which is useful in advanced data visualizations.
4. The visualization again highlights that **West region has the highest sales**, followed by **Central and South regions**.

5. This method demonstrates the ability of Excel to combine **multiple chart types to create complex visualizations**.

Advantages of Overlaid Gridline Visualization

1. Improves comparison of values.
2. Enhances visual clarity.
3. Makes charts more professional.
4. Helps highlight thresholds or benchmarks.

Conclusion

This lab demonstrated how advanced visualization techniques can improve the interpretability of data. By applying overlaid gridline techniques using line charts, formulas, and scatterplots, we created professional and visually appealing charts. These methods allow users to better analyze and compare sales data across regions.